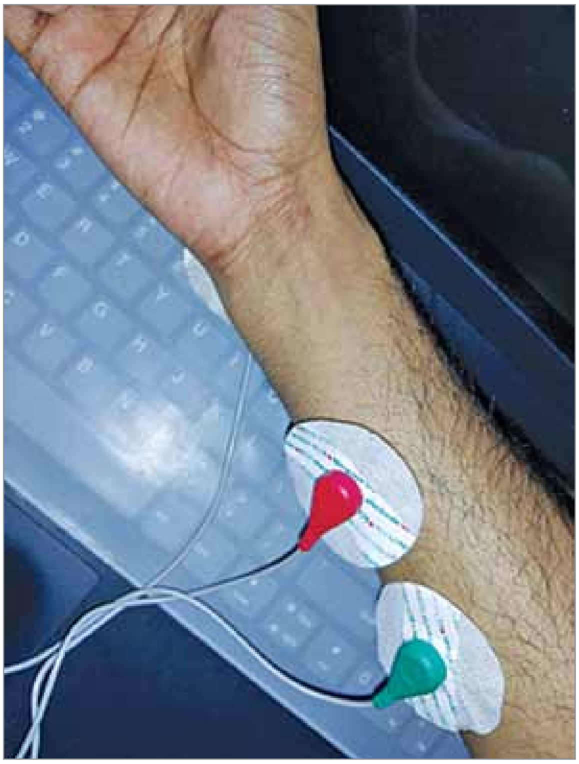

Fig. 11: Electrodes connection for EMG

which is 0 degree to 180 degrees for servo and has values 60 to 800 for the EMG sensor.

Write the mapped value to the servo move position to move the prosthetic hand finger as per signals generated from muscle movement.

## Connection

Connect the components for both transmitter and receiver as shown in Fig. 10 and Fig. 9, respectively. For the EMG transmitter circuit, you need to add the Bluetooth module in the receiver circuit as well.

Please note, you need to configure both the Bluetooth modules for them to pair with each other and transmit/receive data accordingly. To do that, you can follow the instructions available online for setting up master and slave mode of HC05.

To test, attach all fingers to the servo pulley for the prosthetic arm and power the transmitter and receiver-cum-controller for the prosthetic arm. Attach the electrodes to your controlling arm as shown in Fig. 11. The prosthetic arm will mimic your hand's finger movements whenever you close and open your palm.

Congrats! You have just made an extra hand, which can be controlled wirelessly. EFY

Ashwini Kumar Sinha is a technology enthusiast at EFY